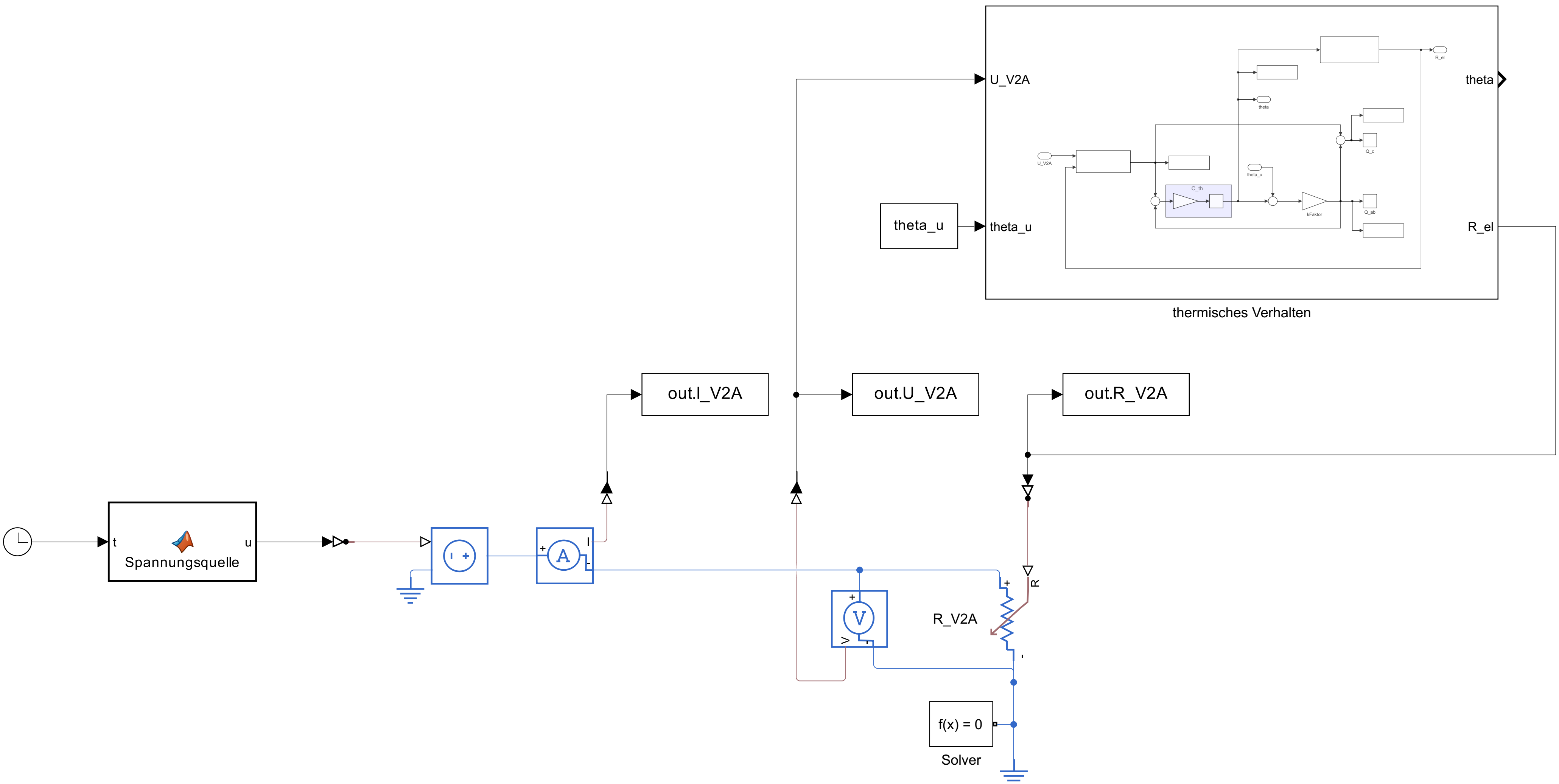
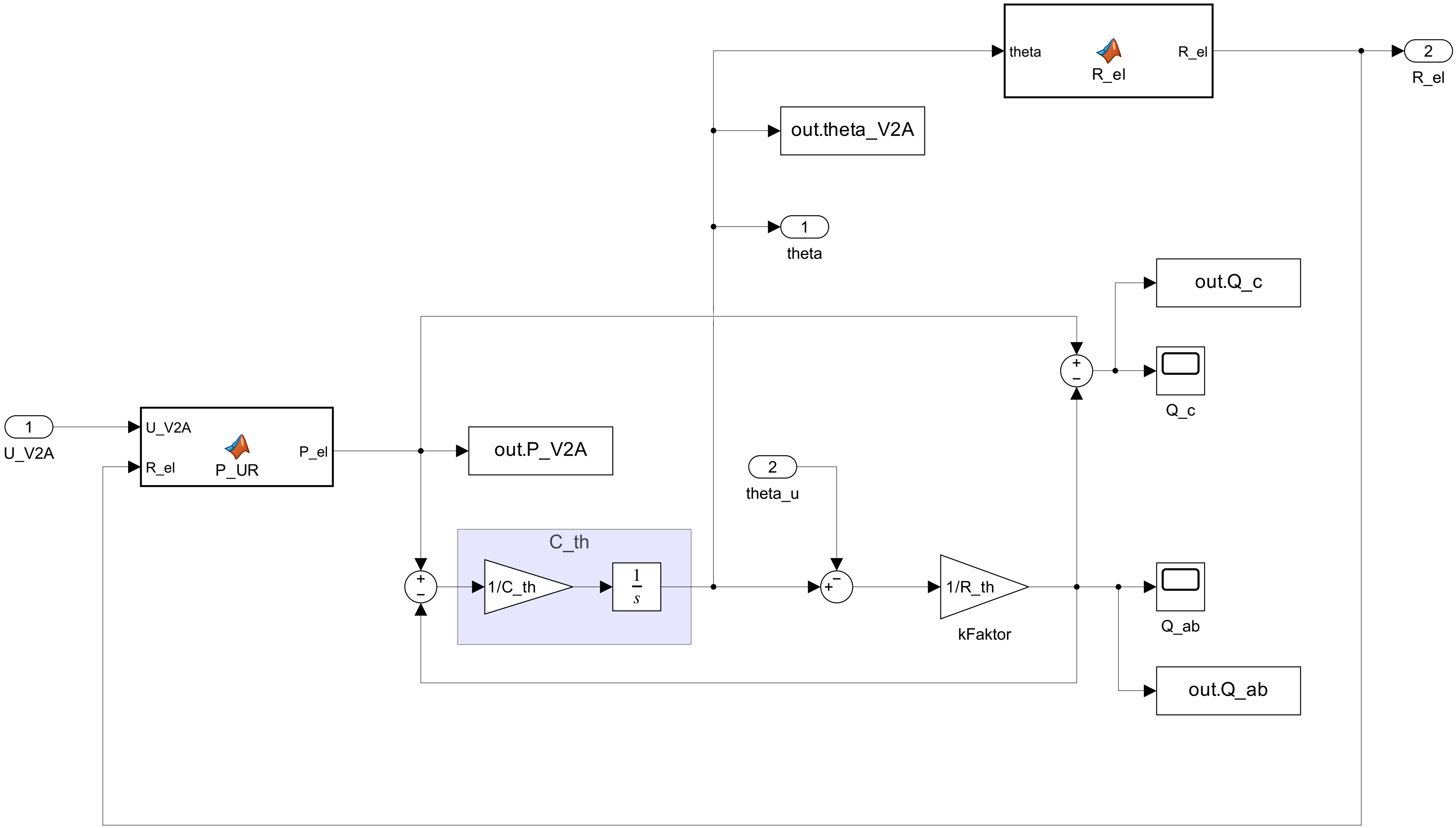

U_V2A
theta
theta_u
theta_u
R_el
thermisches Verhalten
out.I_V2A
out.U_V2A
out.R_V2A
t
Spannungsquelle
u
I
R
R_V2A
f(x) = 0
Solver

```
function u = Spannungsquelle(t, t_sprung, U_sprung)
%#codegen
% Einfacher Gleichspannungssprung von 0 V auf 4 V



    if t < t_sprung
        u = 0.0;
    else
        u = U_sprung;
    end

end
```


theta
R_el
R_el
2
R_el
out.theta_V2A
1
theta
out.Q_c
Q_c
1
U_V2A
U_V2A
R_el
P_UR
P_el
out.P_V2A
2
theta_u
C_th
1/C_th
1/s
1/R_th
kFaktor
Q_ab
out.Q_ab

```
function P_el = P_UR(U_V2A, R_el)
%#codegen
P_el = U_V2A^2 / R_el;
end
```

```
function R_el = R_el(theta, rho_el20, alpha_el, L, b, d)
%#codegen


    % Querschnitt
    A_mm2 = (b * 1000.0) * (d * 1000.0);

    % Widerstand bei 20 °C
    R_20 = rho_el20 * L / A_mm2;

    % Temperaturabhängiger Widerstand
    R_el = R_20 * (1.0 + alpha_el * (theta - 20));

end
```